

SYNOPSIS

At 1100 UTC on 28 May 2003, the container vessel *P&O Nedlloyd Vespucci* and the yacht *Wahkuna* collided in the English Channel in poor visibility. The MAIB was notified of the accident on 29 May, and an investigation started on the same day.

Each vessel had detected the other by radar when at a range of about 6 miles. The container ship was on a course of 255°(T) at a speed of 25 knots. The yacht was on the port bow of the container ship on a course of 012° (C) at a speed of 7.5 knots, and was due to pass about 8 cables ahead of the container ship.

The skipper of the yacht, however, incorrectly estimated from his radar display that *P&O Nedlloyd Vespucci* was passing 1.5 miles ahead of *Wahkuna*, and reduced speed by disengaging his engine. This action, which also resulted in a substantial alteration in the yacht's heading as it lost steerage, put the two vessels on a collision course. The actions of the yacht, the CPA of which now appeared as 2 cables to port on ARPA, concerned and confused the master of *P&O Nedlloyd Vespucci*, but he was reluctant to take any manoeuvring action because he was uncertain of what the yacht would do next.

Minutes later, the vessels collided and the bulbous bow of *P&O Nedlloyd Vespucci* struck the forward part of *Wahkuna*'s hull, demolishing the first 3m of her hull and dismasting her. Despite having sent a lookout to the port bridge wing, the master of the container ship was not aware that a collision had occurred, and continued on passage. The yacht crew had to abandon to a liferaft, where they stayed for 5.5 hours before being rescued.

Several factors contributed to the accident including:

- Misunderstanding by *Wahkuna*'s skipper of which of the Collision Regulations are applicable in fog.
- Over-confidence in the accuracy of ARPA by the master of the container ship
- Acceptance by the master of the container ship of a small passing distance
- The inability of the yacht skipper to use radar effectively
- The failure of both vessels to keep an effective radar lookout
- The high speed of the container vessel
- Poor bridge resource management.

A recommendation has been reiterated to the MCA with regard to the issue of guidance to assist in determining a safe speed in restricted visibility.

Recommendations have also been made to the Royal Yachting Association and the British Marine Federation with a view to improving radar knowledge among yachtsmen. The manufacturer of the locating beacon has been advised to check new and existing beacons for similar faults.

The second officer, who was on watch with the master at the time of the accident, was the holder of an unrestricted second mate's certificate of competency. *P&O Nedlloyd Vespucci* was the fourth vessel of this type he had served on since obtaining his certificate of competency.

The master and most other senior officers work for 4 months on board, followed by 2 months leave.

1.5.2 *Wahkuna*

Wahkuna had a complement of five crew members, which included the owner, who was skipper of the vessel.

He had over 40 years sailing experience and was the holder of an RYA coastal and offshore yachtmaster's certificate of competency.

The remaining crew members had varying degrees of yachting experience and qualifications, ranging from RYA offshore yachtmaster to a day skipper's certificate of competency.

All five crew members belonged to the High Sea Sailing Club of North West London.

1.6 ENVIRONMENTAL CONDITIONS

At the time of the accident, the weather conditions were an easterly wind of force 2 to 3 with very little swell in a sea state of 2. The visibility was very poor with thick fog.

The predicted tidal stream was south-westerly at a rate of approximately 2 knots.

1.7 NARRATIVE OF EVENTS ON *WAHKUNA* (ALL TIMES ARE UTC)

Wahkuna sailed from the port of Dielette, bound for the River Hamble, at 0500 on 28 May 2003. When she sailed the weather conditions were good: wind variable force 1 to 2, sea calm and the visibility 3 to 5 miles.

Once clear of the Channel Islands, a course of 012° to 015° (compass) was set towards the Needles Fairway buoy, and the autopilot engaged. The skipper had the con of the vessel and, along with the remainder of the crew, was stationed in the cockpit. *Wahkuna* was making good a speed of approximately 7.5 knots. Her position was plotted every 20 minutes by one of the crew.

At approximately 0930, the visibility began to deteriorate and, at times, was reduced to approximately 50m. In addition to switching on the automatic fog signal, the skipper instructed the crew to raise the mainsail to increase their visibility both visually and by radar.

Just before 1100, the skipper and one other crew member, both of whom were constantly monitoring the radar which was set to the 6-mile range, detected two targets, about ¼ mile apart, bearing approximately north-east at a range of 6 miles.

The targets were visually tracked, until the range had decreased to 3 miles, when the skipper assumed that *Wahkuna* was on a collision course with the nearest target.

At that point, the skipper, in consultation with the other crew member monitoring the radar, decided to take manual control of the steering and to slow the vessel down to approximately 1 to 2 knots. After a while, it was decided to disengage the main engine to bring the vessel to a stop. At that point both the skipper and other crew member estimated by eye and agreed that the target would pass ahead at a distance of 1.5 miles.

Soon after, one of the crew recalled hearing a fog signal, then saw the bow of a large container vessel looming out of the fog at a distance of approximately 50 to 60 metres to port. In an attempt to avoid a collision, the skipper came hard-to-starboard, and ahead on the main engine.

Unfortunately, his actions were unable to prevent a collision. The container vessel's bulbous bow struck the forward part of *Wahkuna*'s port hull, demolishing the first 3 metres of her bow (**Figure 5**).

Figure 5



Photograph of the damage to *Wahkuna*

As a result of the impact, *Wahkuna* rose up 2 metres on the container vessel's bow wave and slalomed at an angle of 20° to 30° down her starboard side stern-first for a distance of approximately 75 metres, before being dropped back into the sea. The container vessel continued on her passage with *Wahkuna* scraping alongside. Moments later, the container vessel's stern passed, leaving *Wahkuna* in her wake. One of the crew managed to read the word "*Monrovia*" on the vessel's stern, but was unable to make out a name. Another was able to record the GPS position.

The skipper's immediate thoughts were to transmit a "Mayday" by VHF radio but the mainmast, which housed the VHF radio aerial, had been torn away during the impact.

He then opened the forward cabin door and noticed a substantial ingress of water. Closing the door immediately, he instructed the crew to don their lifejackets, which were stored in the main cabin area, and collect some food and water. He then put the main engine astern and began making stern way at a speed of about ½ knot through the water. This reduced the amount of water ingress forward.

A further assessment of the damage was made, but it was soon realised that *Wahkuna* was sinking.

Instructions were given to deploy and inflate the liferaft. After collecting flares and a locating beacon, the crew boarded the liferaft. Soon after, *Wahkuna* sank by the bow.

Once in the liferaft, the crew tried to activate the locating beacon, but the light indicating normal operation failed to illuminate. They were also unable to make any contact by mobile phone and therefore had to wait until they were seen by a passing ship. Some 5½ hours after boarding the liferaft, one of their flares was seen by *Condor Express*, a high-speed ferry operating between Poole, Weymouth and the Channel Islands.

Once recovered on board the ferry, the crew were landed in Guernsey and taken to the local hospital for a routine medical examination. Later on that evening, they sailed to Poole on a return ferry.

1.8 SUBSEQUENT INVESTIGATION

Following the report of the collision to the MAIB, CROSS Jobourg MRCC in France was asked for detailed radar information concerning all vessels transiting the relevant part of the English Channel at the time of the collision, using the position reported and information provided by *Wahkuna*'s crew.

Given the time and the position reported, the tracks of two vessels were interrogated: *Yokohama Senator* and *P&O Nedlloyd Vespucci*, both large container vessels, and both registered in Monrovia (**Annex 1**).

Of the two vessels, only one, *P&O Nedlloyd Vespucci*, was in the vicinity of the position reported at 1100 UTC.

Following enquiries by Ushant Traffic, the master of *P&O Nedlloyd Vespucci* reported that he had "*a radar target coming from his port side on a northerly heading at a speed of 5.5 knots in approximate position 50°10' N, 001° 56' W at 1100 UTC. The target passed on his port quarter at a distance of 2 cables*".

On receipt of the above information, two MAIB inspectors travelled to Hong Kong to interview the master, the officer of the watch and the lookout at the time of the accident, on board *P&O Nedlloyd Vespucci*. During this visit, paint marks of similar colour to the Moody 47 were found on the vessel's starboard bow, supporting the probability that *P&O Nedlloyd Vespucci* was the vessel involved in the collision with *Wahkuna*. Full co-operation was given by the vessel's ship managers, master and crew.

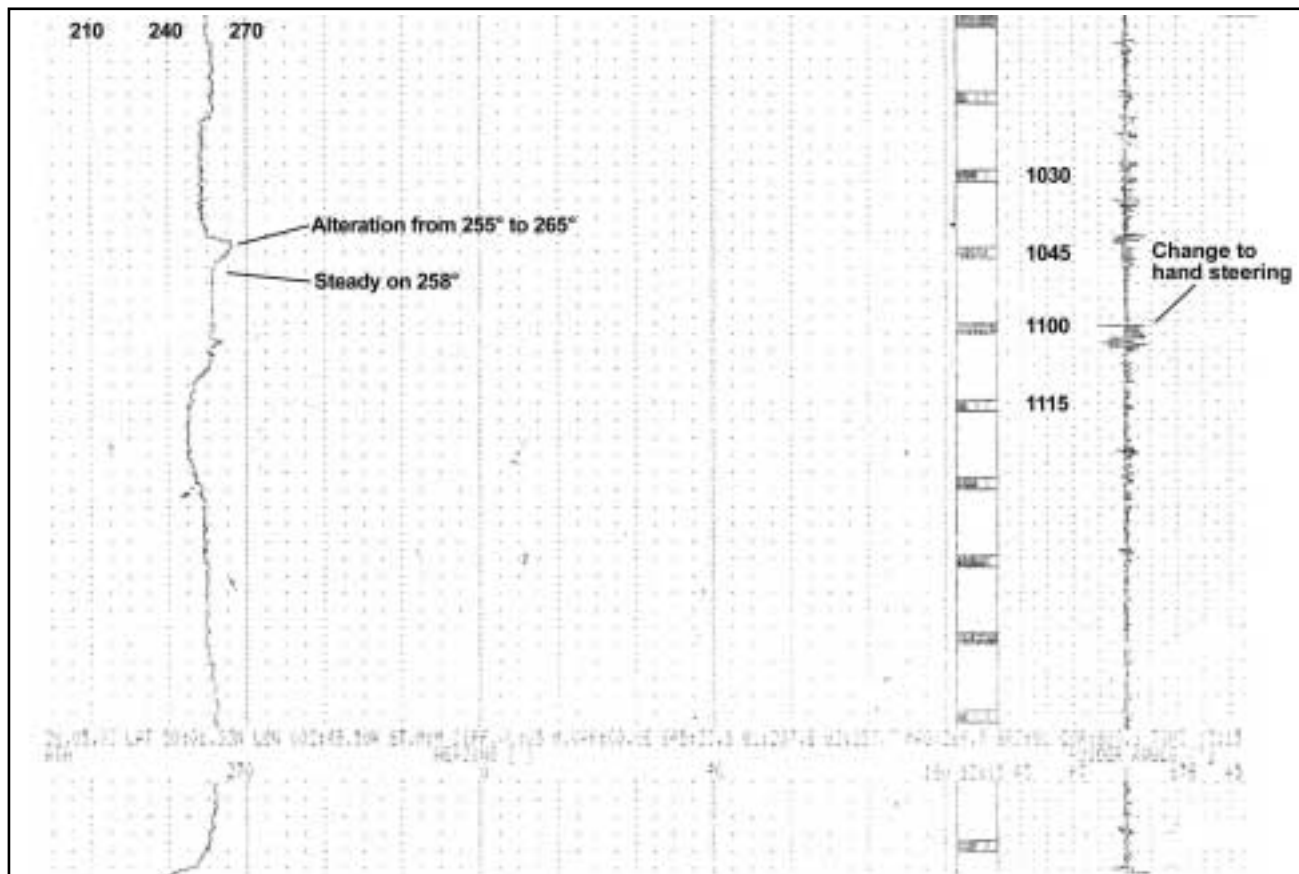
1.9 NARRATIVE OF EVENTS ON *P&O NEDLLOYD VESPUCCI* - (ALL TIMES ARE UTC)

P&O Nedlloyd Vespucci sailed from Antwerp at 2042 on 27 May 2003 for passage to Port Said; she should have sailed at 2000, but was delayed by cargo operations and tidal restrictions. The following day, during passage through the English Channel, visibility started to reduce from about 0800 because of fog. At this time, the ship began sounding fog signals. By about 0900, visibility had reduced to the extent that the bow was not visible from the bridge. The vessel was following a track of 255° in autopilot at a speed of 25 knots.

The master had remained on the bridge since leaving Antwerp. He had accompanied the pilot until he disembarked at 0400, and then remained for the passage past the sunken vessel *Tricolor*, and the transit through the Dover Strait. He then supervised the third officer during his 0800 to 1200 watch. A rating had also been on the bridge during each watch.

Between about 1040 and 1045, a small radar contact was detected at a range of between 4 to 5 miles at an angle of 50° on the port bow; the CPA of the contact was reported to be less than 1 cable. The rating was sent to the port bridge wing and the master altered course about 10° to starboard to increase the CPA. This alteration was maintained for less than 2 minutes before returning to port to a course of 258°. The rating on the port bridge wing never saw or heard the contact.

Soon after, another contact was detected on the port bow at a range of between 5 and 6 miles. By ARPA, its course was northerly and speed was 6.5 knots; the master estimated it would cross 8 cables ahead. However, when about one point on the port bow at a range of between 1.5 and 2 miles, the contact slowed quickly and the ARPA vector indicated that it had also altered course towards the north-east. Its speed by ARPA reduced to zero. This concerned the master,



Extract of *P&O Nedlloyd Vespucci*'s course recorder

who immediately started the second steering motor and ordered the second officer, who was then the OOW, to change to manual steering. This was done, but neither course nor speed was adjusted.

The CPA of the contact was then 2 cables on the port side and the master was uncertain of its intentions. The contact was monitored by the master on the starboard radar display using S-Band radar. The display was set to the 3-mile range scale and, initially, was off-centred to the north-east. The master reported that the contact was never seen to be lost in the radar ground wave or clutter and that its range by ARPA was not less than 2 cables. As the contact passed, the radar was recentred, and the master saw the contact's trail pass 2 cables astern. The time was shortly before 1100. When the contact was between 1.5 to 2 miles astern, its course and speed by ARPA was about 075° at 3 knots. The contact was cancelled from ARPA and the ship continued on passage. The lookout on the port bridge wing did not see or hear anything, and the master was completely unaware that a collision had occurred. An extract of the vessel's course recorder is shown at **Figure 6**, which shows:

- Prior to the alteration of course to starboard between 1040 and 1045, course was 255°.
- The heading reached during the alteration was about 265°.
- The time taken to alter to starboard from 255° to 265° and back to port to 258° was 6 minutes.

- The interval from being steady on 258° until changing to hand steering was 11 minutes.
- While in hand steering, heading oscillated no more than 5° either side of 258°.
- The ship remained in hand steering for about 6 minutes.

1.10 NAVIGATION IN FOG

Advice concerning navigation in fog is contained in *Marine Guidance Note MGN 202 (M&F)*, published by the MCA and entitled *Navigation in Fog*, (**Annex 2**) which states, in part:

The Maritime and Coastguard Agency (MCA) is concerned that a number of casualties to ships have resulted from serious disregard for the basic principles of good seamanship and prudent navigation in bad visibility. Sensible use of radar and other aids to navigation greatly assists the conduct of ships in fog, but these aids have not reduced the need to comply fully with the Collision Regulations: to proceed at a safe speed, pay special attention to good watch-keeping, and navigate with proper caution.

It further gives a brief outline of three casualties in fog, then states:

None of the casualties described led to loss of life, but clearly this was only due to good fortune. In all cases those responsible for the ship's navigation sacrificed seamen for expediency. They failed to recognise the limitations of aids to navigation; or to follow the requirements of the Collision Regulations and the advice of Marine Notices. It is worth stressing that the ships involved were all well-equipped vessels in the charge of men with sound qualifications; it was not skill or experience that was lacking, but the proper seamanlike approach to the situation.

Whatever the pressure on masters to make a quick passage or to meet the wishes of owners, operators, charters or port operators, it does not justify ships and those on board them being put unnecessarily at risk.

The document also stresses the responsibilities of owners; it is the duty of the company to take all reasonable steps to ensure that the ship is operated in a safe manner. The company must have established and implemented an effective safety management system, which includes procedures to ensure safe operation of ships.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of this analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 GENERAL

Fortunately, there were no injuries as a result of this accident. However, there could quite easily have been five fatalities. By good fortune and the last minute evasive action of *Wahkuna*'s skipper which probably reduced the damage sustained, all five crew managed to survive by abandoning their yacht and boarding the liferaft.

Even then, in a busy shipping lane, and having no means of verbal communication, (there was every possibility) they could have been run-down by another vessel, especially in conditions of restricted visibility.

2.3 THE ACCIDENT

Given the position and time of collision, the identification of the container vessel's country of registry by the crew of *Wahkuna*, the radar information provided by CROSS Jobourg, the account of the master of *P&O Nedlloyd Vespucci*, and the blue paint on the vessel's starboard bow, the MAIB is certain that the vessel in collision with *Wahkuna* was *P&O Nedlloyd Vespucci*. It is considered that *Wahkuna* was the radar target, which the master first detected at a range of between 5 and 6 miles and estimated would pass about 8 cables ahead.

At that time, *Wahkuna* was on a compass course of about 012° (about 009° true) at a speed of about 7.5 knots, which was indicated by the ARPA on board *P&O Nedlloyd Vespucci*. The master's observation that the radar contact suddenly slowed when at a range of between 1.5 and 2 miles also correlates with the actions taken by the yacht's skipper.

After *Wahkuna* had started to reduce speed, however, the ARPA information used by the master could not have been accurate. For the vessels to collide, the CPA must have been zero, not 2 cables. Had he monitored the radar echo, he would have recognised that a collision was imminent. Subsequently, the master was unaware of the yacht's predicament, as he saw *Wahkuna* clearly painting a radar 1.5 to 2 miles astern. He had no way of knowing that the course and speed of the target displayed by ARPA as 075 degrees at 3 knots was probably due to inaccuracies of the ARPA, and later the very slow astern movement of the yacht to try and stay afloat, rather than her continuing on passage. His decision to continue was, therefore, understandable. *P&O Nedlloyd Vespucci*

was in position 50°11'10"N, 001°54'10" making good 27 knots through the water on a course of 257.6°. From this it is calculated that she passed within 1 cable of the position of the collision at about 1057.

An estimation of the tracks of *P&O Nedlloyd Vespucci* before the collision is shown at **Figure 7**. This has been constructed using information from CROSS Jobourg, *P&O Nedlloyd Vespucci*'s course recorder, and personal accounts of those on board both vessels. The information provided by CROSS Jobourg for *P&O Nedlloyd Vespucci* at 10:35:09, 10:45:09 and 10:55:09, showing that the vessel's heading was 254.1°, 261.7° and 257.6° respectively, correlates with the vessel's heading shown on her course recorder. The headings recorded during this period reflect the manoeuvring from 255° to 265° and to 258°.

As it is known that the manoeuvre spanned 10:45:09, and that the course recorder shows that *P&O Nedlloyd Vespucci* was steady on a course of 258° for about 11 minutes after the alteration until changing to hand steering, the manoeuvre must have been completed by 1046 at the latest. This would have been at a distance of 4.95 miles from the position of the collision. The probability that *P&O Nedlloyd Vespucci* was altering from 265° to 258° at 10:45:09 is also supported by the fact that the course made good between 10:35:09 and 10:45:09 was 260°, compared to 257° between 10:45:09 and 10:55:09. This indicates that most of the manoeuvre was completed in the first period. However, as *P&O Nedlloyd Vespucci* was unlikely to have been steady on a course of 258° much before 1046, even allowing a time delay for the shore radar to process her changing course, it follows that hand steering was probably selected at a closer range to *Wahkuna* than recalled by her bridge team.

2.4 ACTION TAKEN

2.4.1 *P&O Nedlloyd Vespucci*

When the bridge team first detected *Wahkuna*, the master was content with the vessel crossing 8 cables ahead. He did not consider this to be a close quarters situation even though given both vessel's courses and speeds, the CPA would have been between 2 to 3 cables to starboard. Consequently, the master took no action.

After the yacht had reduced speed and the CPA by ARPA had changed to 2 cables to port, the master became so concerned by *Wahkuna*'s actions, which were unexpected, that he considered it necessary to change to manual steering and put the OOW on the helm. However, he was confused by the unexpected actions of the yacht to the extent that he was reluctant to take any manoeuvring action because of the uncertainty of what the yacht would do next.

This raises the issue of what constitutes a close quarters situation, and the action that should be taken.

A N Cockcroft and J N F Laeijer state: *The distance at which a close quarters situation first applies has not been defined in miles, and is not likely to be, as it will depend upon a number of factors. The 1972 Conference (IMO Revision of the Collision Regulations) considered the possibility of specifying the distance at which it would begin to apply but after lengthy discussions it was decided that this distance could not be quantified.*

On the other hand, the Seafarers' International Research Centre (SIRC) in a paper dealing with near miss encounters in the Dover Strait (Belcher P (2002) *"Overtaking in the Dover Strait, an analysis of near miss encounters"*) states: *from a review of literature on ships' domains (Fuji and Tanka, 1971, Goodwin, 1977, Coldwell, 1983, Zhao, et al., 1993) it has been found that the domain required for a ship in congested waters can be approximated to a circular space with a radius of 8 cables. It also states: it might be argued that a criterion of a minimum passing distance of 8 cables is too stringent a measure for such a busy area. However, a passing distance of 3 cables or less, is on anyone's measure, a very dangerous occurrence that could lead to a collision with only a very slight change in circumstances.*

In this case, where the vessel was transiting in thick fog at high speed, the master's acceptance of such a small passing distance was inappropriate. Given the manoeuvrability of his vessel, and the light traffic density, both a bold alteration to starboard within the 1.5 mile to 2 miles sea room available in accordance with Rules 2(a) and 19(d), or a reduction in speed in accordance with Rule 19(e) could have easily been made (**Annex 4**). Either course of action would have prevented the collision, but neither was taken.

Fleet Instructions, which were on board *P&O Nedlloyd Vespucci* in accordance with ISM Code requirements, stated that close quarters situations should be avoided by early and substantial course alterations and that in open sea, plenty of room should be given to all traffic. In addition, they stated that a safe CPA was considered 1 mile or more in the open sea, but that the safest CPA should always be obtained. Although there was no specific guidance regarding CPAs in shipping lanes, the ship was not in confined waters and there were few other vessels in the vicinity, so there was no reason why the master could not have complied with his own standing orders and the Fleet Instructions.

Several factors probably contributed to the master's acceptance of such a small CPA and lack of action, including; over-confidence in the accuracy of ARPA, the short period of time available in which to assess and react to the situation, and fatigue. Although having rested in Antwerp, the master had been awake from 1600 on 27 May, and had been on the bridge for 14 hours at the time of the accident. Being on the bridge for such a long period would inevitably have been tiring, which might have reduced his alertness to the potential consequences of the situation.

2.4.2 *Wahkuna*

The skipper of *Wahkuna* was under the mistaken assumption that he was "the stand-on vessel". There is no such thing in the Collision Regulations as a stand-on vessel in restricted visibility. Section II of the Collision Regulations (Rules 11 - 18) apply to the "Conduct of Vessels in sight of one another". Section III (Rule 19) governs the "Conduct of Vessels in Restricted Visibility". Rule 19d states "*A vessel which detects by radar alone the presence of another vessel shall determine if a close-quarters situation is developing and/or risk of collision exists. If so, she shall take avoiding action in ample time, provided that when such action consists of an alteration of course, so far as possible the following shall be avoided:*

- (i) *an alteration of course to port for a vessel forward of the beam, other than for a vessel being overtaken;*
- (ii) *an alteration of course towards a vessel abeam of abaft the beam."*

Thus the onus in fog is for both vessels to take "avoiding action in ample time" to avoid a close quarters situation developing.

The skipper of *Wahkuna* had slowed down after estimating by eye on his radar display, that *P&O Nedlloyd Vespucci* was passing about 1.5 miles ahead. As the vessels eventually collided without *P&O Nedlloyd Vespucci* altering either course or speed, it is evident that had the yacht not reduced speed, she would have passed safely ahead of the container ship.

It is unlikely that *Wahkuna* completely stopped in the water after disengaging her engine due to her momentum, and the light wind acting upon her raised sail. The fact that the crew of the yacht saw the bow of the container ship off the port side also indicates that the yacht had lost steerage, and had substantially altered her heading towards the south-east.

The actions taken by the yacht were based on an inaccurate assessment of the situation by radar and served to confuse the bridge team on *P&O Nedlloyd Vespucci*, as well as putting the two vessels on a collision course. Had the skipper been able to make full use of his radar, a more accurate assessment would have been possible, and it would have been apparent that a substantial alteration of course to starboard in accordance with Rule 19(d) would have been more appropriate. This would not only have resulted in the vessels passing at a safe distance, it would also have shown clear intent and have been readily apparent to the bridge team on *P&O Nedlloyd Vespucci*.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES

The following are the safety issues identified by the investigation. They are not listed in any order of priority.

1. The acceptance of a 2 cable passing distance when transiting at high speed in thick fog was inappropriate and contrary to the ship manager's Fleet Instructions [2.4.1]
2. The factors contributing to the master's acceptance of this small passing distance probably included: over-confidence in the accuracy of ARPA, the short period of time available in which to assess and react to the situation, and fatigue. [2.4.1]
3. The master of *P&O Nedlloyd Vespucci*'s alertness to the potential consequences of the situation might have been reduced after spending 14 hours on the bridge overnight. [2.4.1]
4. The action taken by the skipper of *Wahkuna*, which put the vessels on a collision course, was not expected by the master of *P&O Nedlloyd Vespucci* and confused him. [2.4.1 and 2.4.2]
5. There was considerable scope for inaccuracy in the information displayed by ARPA. [2.5.1]
6. The radar information on *P&O Nedlloyd Vespucci* was ground-stabilised, which can be misleading when assessing target aspect. [2.5.1]
7. Neither the master of *P&O Nedlloyd Vespucci* nor the skipper of *Wahkuna* kept an appropriate radar lookout as the vessels approached each other, and both made decisions based on scanty radar information. [2.5.2]
8. It is imperative, whenever radar equipment is fitted on board any type of vessel, watchkeepers are fully versed and trained in its use, capabilities, and limitations. [2.5.3]
9. The lack of a helmsman meant that the second officer had to undertake this duty, rather than enhancing the radar lookout and assisting the master in his assessment of the situation. [2.6]
10. At 25 knots, the master had less than 5 minutes to assess the situation and react to the yacht's unexpected action using potentially inaccurate data, and could only have stopped after 2.4 miles, which was beyond the yacht's position at the time. [2.7]
11. Had the locating beacon not been corroded and had functioned correctly, it might have reduced the time the yacht's crew had to spend in the liferaft. [2.9]